Imagine you are an alien (who knows mathematics and can think)

Here is your task:

* I will give you an incomplete sentence. You have to predict the next word.
Remember: you don't know language.
How will you do this? *

Your effort will take you ?

↳ Student answers?

* Let's say the alien reads all web data + books.
The alien tries to estimate the probability of what can come next.

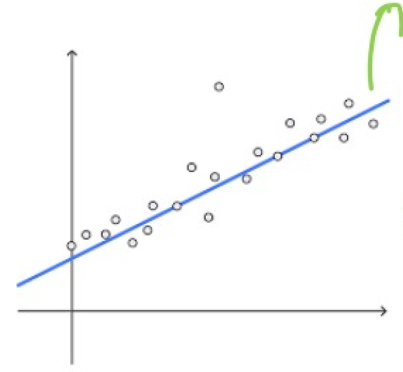
Your effort will take you 40000 options
→ 40000 * 40000 options
→ 40000 * 40000 * 40000 options → 1.6 trillion

more options than the number of particles in the universe

20

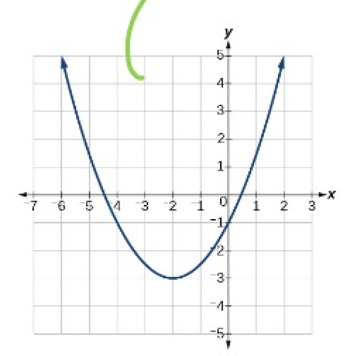
What if I make a model which estimate the probabilities with which the next word can appear?

↳ Such a model is called as a large language model



$y = ax + b$

→

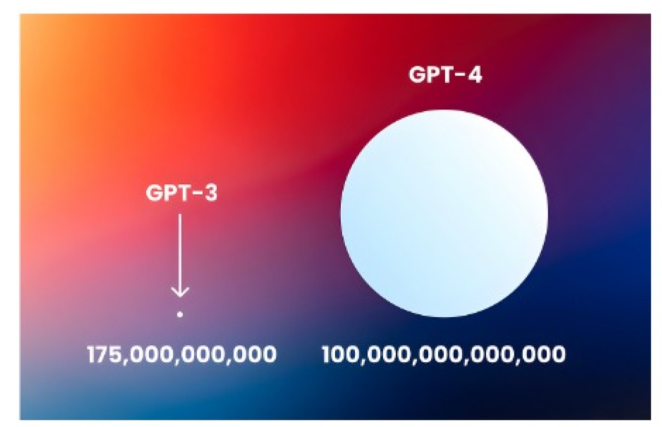


$ax^2 + bx + c$

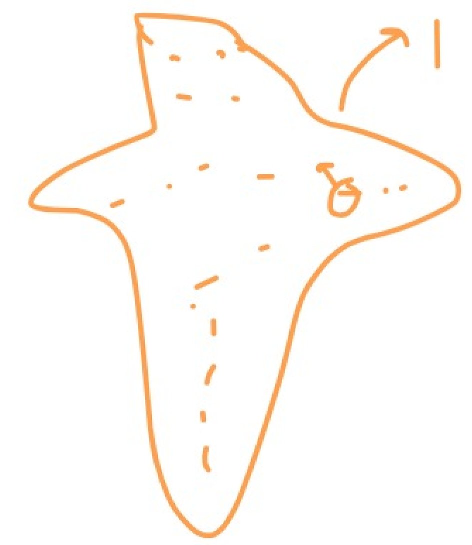
→

Sequence of words → LLM → Next token

↳ language as a by-product



We will make a model with billions of parameters



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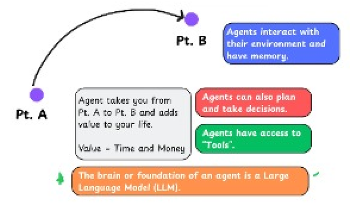
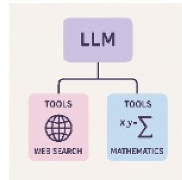
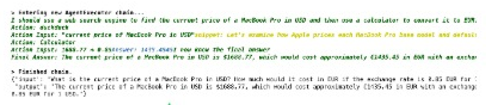
The theory behind agents

Let's first see an agent in action!







What just happened here?

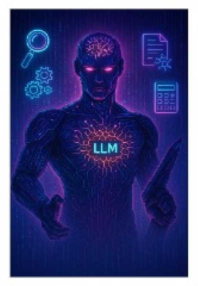
* Did we see something magical?

Once we understand it, it will no longer be magical.

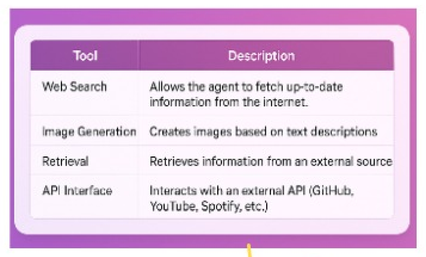
* You need to understand 2 things to really understand the code.

Tools

ReAct Framework.



Tools are the weapons which an LLM recieves to interact with its environment.



Tool	Description
Web Search	Allows the agent to fetch up-to-date information from the internet.
Image Generation	Creates images based on text descriptions
Retrieval	Retrieves information from an external source
API Interface	Interacts with an external API (GitHub, YouTube, Spotify, etc.)

Custom tool

ReAct (Reasoning + Acting)

① A good tool complements the power of an LLM

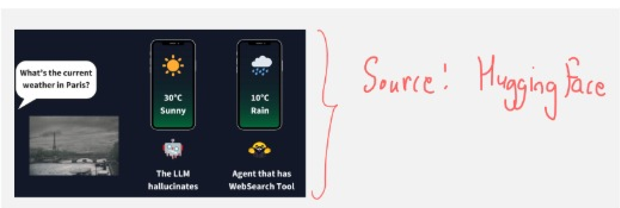
② If LLM struggles with maths

give it the calculator tool

③ If LLM cannot find latest info, provide it with a web search tool.

Weather

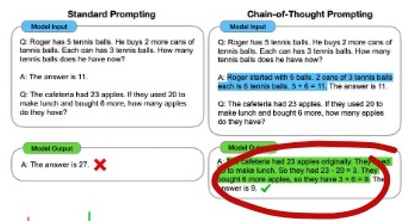
weather api



Source: Hugging Face

Here is how ReAct was born:

(1) Chain-of-Thought (CoT) prompting:



However, CoT lacks access to external world. Hence, it cannot act.

This technique proved that LLMs are capable of generating reasoning traces (or thoughts)

(2) ReAct builds on top of CoT.

ReAct = LLM + Reasoning, Acting

Price needs to be converted to Euro

I have to check Macbook prices

Thought

Let me use calculator tool

Human feedback

Observation

Action

Let me do web search

Price seems to be in dollars

price in Euro. Satisfied

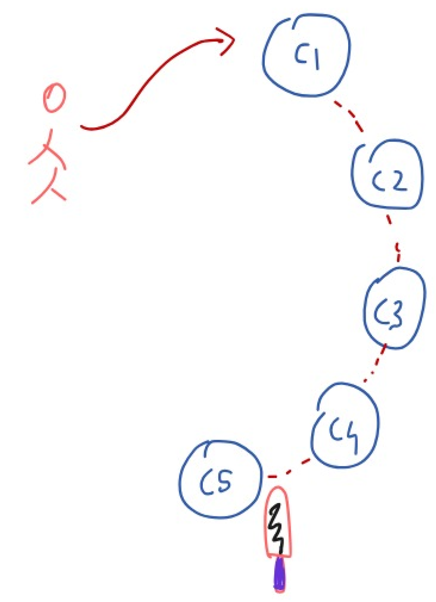
Agents are LLMs equipped with tools.

Using the ReAct framework, agents interact with external environment (eg: websites) to add additional information into their reasoning.

Consider a kid

Thought - Action

Observation



5 chairs

Kid has to find chair with candy

My takeaway after understanding how tools work

Prompt engineering is very important!

How do tools work?

LLMs can only receive input texts and generate output. They have no way to call tools on their own.

Prompt

We need to tell the LLM!

- Hey LLM, this tool exists
- Here's how to use it

text → call_weather_tool

LLM

response

Here's what happens next:

① What is the weather in Paris? → ② LLM → ③ Generates text that represents tool call → ④ Agent reads this response

Recognizes that weather tool can be used

"call weather_tool ('Paris')"

⑤ Agent executes tool on LLM's behalf

⑥ Agents get weather data

⑦ LLM uses this weather data for further responses

The user feels that the LLM has directly interacted with the tool, but actually, the agent has handled the entire workflow in the background

⑥ How do we give tools to an LLM?

We provide this information in the system prompt!

```
system_message="""You are an AI assistant designed to help users efficiently and accurately. Your
primary goal is to provide helpful, precise, and clear responses.

You have access to the following tools:
{tools_description}
"""
```

Source: Hugging Face

⑦ For this to work, we have to be very precise and accurate about:

- What the tool does?
- What exact inputs it expects?